

A Homogeneous Rearrangement-Invariant Space That Is Not Ultrasymmetric

fa_banach_001 / agent_lane_00

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Status: candidate full negative resolution of Remark 2.14.

1 The question and the answer

Boza, Křepela, and Soria ask whether every p -homogeneous rearrangement-invariant (r.i.) space on $(0, \infty)$ must be ultrasymmetric for the fundamental function $\varphi(t) = t^{1/p}$ [1, Remark 2.14].

$i \in \{0, 1\}$, [2, Proposition III.1.11] provides a constant $C > 0$ such that $\|D_r f\|_X \leq C\|f\|_X$, for all $r > 0$ and $f \in X$. By Proposition 2.7, $\|\cdot\|_X$ is equivalent to a p -homogeneous r.i. norm. Proving the rest is simple. $\square$

Remark 2.14. An interesting open problem is whether Theorem 2.13 is actually a characterization of the p -homogeneity condition. Recall that an *ultrasymmetric space* X is defined as an r.i. interpolation space for the couple $(\Lambda_\varphi, M_\varphi)$, formed by the Lorentz and the Marcinkiewicz spaces with fundamental function φ (cf. [11]). Now, the question is the following: is it true that every p -homogeneous r.i. space X is ultrasymmetric for $\varphi(t) = t^{1/p}$? The previous theorem shows that the reverse implication is true, up to an equivalent renorming. Notice that [11] provides a characterization of ultrasymmetry in terms of existence of a certain r.i. space with respect to the measure $\frac{dt}{t}$. A possible way of tackling the suggested problem could therefore be to show that mere p -homogeneity suffices to construct such a space. Whether this is possible however remains an open question.

The answer is no, already for $p = 2$. For a measurable function f , let f^* be the nonincreasing rearrangement of $|f|$ and put

$$f^{**}(t) = \frac{1}{t} \int_0^t f^*(u) du, \quad G_f(s) = e^{s/2} f^{**}(e^s) \quad (s \in \mathbb{R}).$$

For a nonnegative measurable g on $\mathbb{R}$, define the translation-invariant Morrey functional

$$\mathfrak{M}(g) = \sup_I |I|^{-1/2} \int_I g(s) ds, \tag{1}$$

where the supremum is over all bounded nondegenerate intervals $I \subset \mathbb{R}$.

Theorem 1. *The functional*

$$\|f\|_X := \mathfrak{M}(G_f) \tag{2}$$

is a rearrangement-invariant Banach function norm on $(0, \infty)$ and is exactly 2-homogeneous:

$$\|D_r f\|_X = r^{-1/2} \|f\|_X \quad (r > 0).$$

Its fundamental function is a constant multiple of $t^{1/2}$, but X is not ultrasymmetric. Consequently the question in Remark 2.14 of [1] has a negative answer.

2 Proof intuition

The change of variables $t = e^s$ turns dilation into translation. Exact 2-homogeneity therefore requires only a translation-invariant norm of the logarithmic profile G_f . Ultrasymmetry is stronger: Pustynnik's characterization requires a fully rearrangement-invariant parameter norm on the logarithmic line [2].

The Morrey functional in (1) remembers where mass sits. We construct two profiles from the same multiset of gaps. Their decreasing rearrangements are uniformly comparable, but one ordering groups many kernels in a cluster while the other disperses them into isolated pairs. Every r.i. parameter norm must treat the profiles comparably, whereas $\mathfrak{M}$ makes their norms differ by order $\sqrt{m}$.

3 The norm and its homogeneity

Lemma 1. *The functional in (2) satisfies the Banach-function-norm axioms, is r.i., and obeys*

$$G_{D_r f}(s) = r^{-1/2} G_f(s + \log r). \quad (3)$$

Moreover, for a measurable set E of measure a ,

$$\|\chi_E\|_X = a^{1/2} \mathfrak{M}(k), \quad k(s) := e^{-|s|/2}. \quad (4)$$

Proof. The functional depends only on f^* , so it is rearrangement invariant. The standard subadditivity

$$(f + g)^{**}(t) \leq f^{**}(t) + g^{**}(t)$$

gives the triangle inequality, and lattice monotonicity is immediate. Definiteness and scalar homogeneity are clear.

For the Fatou axiom, suppose $0 \leq f_n \uparrow f$. The identity

$$t f_n^{**}(t) = \sup_{|A| \leq t} \int_A f_n$$

and monotone convergence show that $f_n^{**}(t) \uparrow f^{**}(t)$ for every $t > 0$. Monotone convergence on each interval, followed by the interval supremum, gives $\|f_n\|_X \uparrow \|f\|_X$.

If $f = \chi_E$ and $|E| = a$, direct calculation gives

$$G_f(s) = a^{1/2} k(s - \log a).$$

Since $k \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$, $0 < \mathfrak{M}(k) < \infty$, proving the finite-characteristic-function axiom and (4).

It remains to verify local integral control. Monotonicity of $f^{**}(t)$ and of $t f^{**}(t)$ implies, for $|u| \leq 1$,

$$G_f(s + u) \geq e^{-|u|/2} G_f(s). \quad (5)$$

Consequently

$$\mathfrak{M}(G_f) \geq 2^{-1/2} \int_{s-1}^{s+1} G_f(v) dv \geq \sqrt{2} e^{-1/2} G_f(s).$$

For any measurable E with $|E| = a < \infty$, the Hardy–Littlewood inequality now yields

$$\int_E |f| \leq \int_0^a f^*(t) dt = a f^{**}(a) = a^{1/2} G_f(\log a) \leq \frac{e^{1/2}}{\sqrt{2}} a^{1/2} \|f\|_X.$$

This proves the remaining Banach-function-norm axiom. The usual theorem for Banach function norms then supplies completeness.

Finally, $(D_r f)^{**}(t) = f^{**}(rt)$, which proves (3). Translation invariance of (1) gives $\|D_r f\|_X = r^{-1/2} \|f\|_X$. $\square$

4 Profiles with identical gap data

Let $S = \{s_1 < \dots < s_N\}$ be a finite set and define the decreasing step function

$$f_S(t) = \sum_{i=1}^N e^{-s_i/2} \chi_{(0, e^{s_i}]}(t). \quad (6)$$

Thus $f_S^* = f_S$. Introduce the two logarithmic profiles

$$H_S(s) := e^{s/2} f_S^*(e^s), \quad G_S(s) := e^{s/2} f_S^{**}(e^s).$$

A direct calculation gives

$$H_S(s) = \sum_{s_i \geq s} e^{(s-s_i)/2}, \quad G_S(s) = \sum_{i=1}^N k(s - s_i). \quad (7)$$

Assume all consecutive gaps are at least a fixed $L > 0$. For $s \leq s_N$, let s_j be the first center at or to the right of s and set

$$Q_S(s) = e^{(s-s_j)/2},$$

put $Q_S(s) = 0$ for $s > s_N$. The minimum-gap assumption and a geometric series give

$$Q_S \leq H_S \leq C_L Q_S, \quad C_L = (1 - e^{-L/2})^{-1}. \quad (8)$$

If the gaps are $d_i = s_{i+1} - s_i$, then for $0 < \lambda < 1$,

$$|\{Q_S > \lambda\}| = 2 \log(1/\lambda) + \sum_{i=1}^{N-1} \min\{d_i, 2 \log(1/\lambda)\}. \quad (9)$$

Indeed, the leftmost half-line contributes $2 \log(1/\lambda)$, while the interval of length d_i immediately preceding s_{i+1} contributes the minimum appearing in the sum.

It follows that the distribution of Q_S depends only on the *multiset* of gaps, not their order. Therefore, whenever S and T have the same gap multiset and minimum gap at least L , (8) implies

$$C_L^{-1} H_T^* \leq H_S^* \leq C_L H_T^*. \quad (10)$$

5 Clustering versus dispersion

Fix $L \geq 4$. For each integer $m \geq 2$, put $R_m = Lm^2$. Let S_m have the ordered gap list

$$\underbrace{L, \dots, L}_{m \text{ times}}, \underbrace{R_m, \dots, R_m}_{m \text{ times}}, \quad (11)$$

and let T_m have the alternating gap list

$$L, R_m, L, R_m, \dots, L, R_m. \quad (12)$$

Both sets contain $2m + 1$ centers and have the same gap multiset.

Lemma 2. *There are constants $c_L, C'_L > 0$, independent of m , such that*

$$\mathfrak{M}(G_{S_m}) \geq c_L \sqrt{m}, \quad \mathfrak{M}(G_{T_m}) \leq C'_L.$$

Proof. In S_m , the first $m + 1$ centers have spacing L . The interval extending one unit beyond the two endpoints of this cluster has length $mL + 2$ and captures at least $c_0 := \int_{-1}^1 k(u) du > 0$ from each of those kernels. Hence

$$\mathfrak{M}(G_{S_m}) \geq \frac{c_0(m+1)}{\sqrt{mL+2}} \geq c_L \sqrt{m}.$$

For the dispersed set, first note two elementary estimates valid for any finite set of centers with minimum gap L . The geometric decay of k gives

$$\sup_s \sum_i k(s - s_i) \leq A_L, \quad (13)$$

and, for every interval I ,

$$\int_I \sum_i k(s - s_i) ds \leq 4\#(S \cap I) + B_L. \quad (14)$$

For (14), the centers inside I contribute at most $\int_{\mathbb{R}} k = 4$ each, while the centers outside either endpoint form two geometrically summable tails.

The alternating construction gives, for $\ell = |I|$,

$$\#(T_m \cap I) \leq \min\{2m + 1, 2\ell/R_m + 5\}.$$

If $\ell \leq 1$, (13) gives $\ell^{-1/2} \int_I G_{T_m} \leq A_L$. If $1 < \ell \leq mR_m$, (14) and the local count give

$$\ell^{-1/2} \int_I G_{T_m} \leq C_L \left(\frac{\sqrt{\ell}}{R_m} + \frac{1}{\sqrt{\ell}} \right) \leq C_L \left(\sqrt{\frac{m}{R_m}} + 1 \right).$$

For $\ell > mR_m$, use the total count instead to obtain

$$\ell^{-1/2} \int_I G_{T_m} \leq C_L \frac{m}{\sqrt{mR_m}} + C_L \leq C'_L.$$

Since $R_m = Lm^2$, all bounds are uniform in m . □

6 Failure of ultrasymmetry

Completion of the proof of Theorem 1. Suppose, toward a contradiction, that X is ultrasymmetric. By (4), its fundamental function is equivalent (indeed equal up to a constant) to $t^{1/2}$. Pustynnik's characterization of ultrasymmetric spaces [2, Theorems 2.1–2.2] then supplies an r.i. space E on $((0, \infty), dt/t)$ such that

$$\|f\|_X \asymp \|t^{1/2} f^*(t)\|_E. \quad (15)$$

After $t = e^s$, the parameter space is rearrangement invariant on $(\mathbb{R}, ds)$ and the function on the right of (15) is H_S for $f = f_S$.

The gap multisets of S_m and T_m agree. Hence (10) and the lattice and rearrangement-invariance properties of E imply

$$\|H_{S_m}\|_E \asymp_L \|H_{T_m}\|_E$$

with constants independent of m . Equation (15) would therefore make $\|f_{S_m}\|_X / \|f_{T_m}\|_X$ uniformly bounded. But (2), (7), and Lemma 2 give

$$\frac{\|f_{S_m}\|_X}{\|f_{T_m}\|_X} = \frac{\mathfrak{M}(G_{S_m})}{\mathfrak{M}(G_{T_m})} \geq c'_L \sqrt{m} \rightarrow \infty.$$

This contradiction proves that X is not ultrasymmetric. □

Verification, novelty, and scope

The counterexample is an ordinary Banach r.i. space, not merely a quasi-Banach space. It has exact homogeneity for every real dilation, not only dyadic dilations. The obstruction is norm-equivalence invariant: the contradiction uses the representation of every ultrasymmetric space up to equivalent norm. Only $p = 2$ is needed to refute the universal question; analogous asymmetric exponential kernels should treat other $1 < p < \infty$, but that extension is not claimed here.

The bounded novelty search covered the exact source question, the article title and identifier, homogeneous r.i. spaces, ultrasymmetric spaces, and Pustyl'nik's representation. The source was updated in January 2026 and published in July 2026 with the question still present. No later answer was located. Novelty is therefore provisional pending expert review.

References

- [1] S. Boza, M. Křepela, and J. Soria, *Rearrangement-invariant norms commuting with dilations*, Journal of Mathematical Analysis and Applications 559 (2026), Article 130469; arXiv:2501.15565v2.
- [2] E. Pustyl'nik, *Ultrasymmetric Spaces*, Journal of the London Mathematical Society 68 (2003), 165–182.